

# Two Cubic Perron Numbers with Different Minimal Matrix Sizes: A Disproof of Conjecture 00000009611

GitHub: gaochengzhecpu

## Abstract

The source asserts that the minimal size of a nonnegative integer matrix realising a Perron number  $\lambda$  as the entropy of a mixing subshift of finite type is a function  $f(d)$  of the algebraic degree  $d$  of  $\lambda$ , and that  $f(d) = d$  exactly when the minimal polynomial is sign-alternating. We exhibit two Perron numbers of degree 3: the real root  $x$  of  $X^3 - X - 1$ , whose minimal size is 3, and the real root  $y$  of  $X^3 + X^2 - 3X - 4$ , whose minimal size is 4. So no such function  $f$  exists. The minimal polynomial of  $x$  is not sign-alternating although its minimal size equals its degree, so the stated criterion fails as well. All of this is verified in Lean with Mathlib.

## Reading of the source

Let  $A$  be an  $n \times n$  matrix with entries in  $\mathbb{Z}_{\geq 0}$ ,  $n \geq 1$ . It is *primitive* if some power  $A^k$ ,  $k \geq 1$ , has all entries positive; the subshift of finite type  $X_A$  is mixing exactly in this case. A primitive matrix has a unique eigenvalue with an entrywise positive eigenvector, its Perron eigenvalue  $\lambda_A$ ; this is its spectral radius, and the entropy of  $X_A$  is  $\log \lambda_A$ . A *Perron number* is a real algebraic integer  $\lambda \geq 1$  that is strictly larger than the modulus of each of its other conjugates. By Lind's theorem the Perron numbers are exactly the numbers  $\lambda_A$ .

We say that  $\lambda$  is *realised in size  $n$*  if  $\lambda = \lambda_A$  for a primitive  $n \times n$  matrix  $A$  over  $\mathbb{Z}_{\geq 0}$ , and we write  $m(\lambda)$  for the least such  $n$ . The degree  $d(\lambda)$  is the degree of the minimal polynomial of  $\lambda$  over  $\mathbb{Q}$ . The source makes two claims.

(C1) There is a function  $f$  with  $m(\lambda) = f(d(\lambda))$  for every Perron number  $\lambda$ .

(C2)  $f(d) = d$  if and only if the coefficients of the minimal polynomial of  $\lambda$  are sign-alternating. Since the right-hand side depends on  $\lambda$ , we read this for each  $\lambda$  separately:  $m(\lambda) = d(\lambda)$  if and only if the minimal polynomial of  $\lambda$  is sign-alternating.

The English and the Chinese text agree. We refute (C1), and we refute the implication from left to right in (C2).

## Statement

**Theorem 1.** *Let  $x$  be the real root of  $X^3 - X - 1$  and let  $y$  be the real root of  $X^3 + X^2 - 3X - 4$  in the interval  $[9/5, 2]$ . Then*

1.  $x$  and  $y$  are Perron numbers of degree 3;
2.  $m(x) = 3$ ;
3.  $m(y) = 4$ . Moreover  $y$  is not an eigenvalue of any matrix of size at most 3 with entries in  $\mathbb{Z}_{\geq 0}$ , primitive or not.

**Corollary 1.** *Claim (C1) is false: a function  $f$  would satisfy  $f(3) = 3$  and  $f(3) = 4$ . The implication  $m(\lambda) = d(\lambda) \Rightarrow$  sign-alternating of (C2) is false:  $m(x) = d(x) = 3$ , and the coefficients of  $X^1$  and  $X^0$  in  $X^3 - X - 1$  are both  $-1$ .*

Numerically  $x \approx 1.3247$  and  $y \approx 1.8312$ .

## Proof

**Lemma 1.** *Let  $A$  be an  $n \times n$  matrix over  $\mathbb{Z}_{\geq 0}$  and let  $v \in \mathbb{R}^n$  have positive entries with  $Av = \lambda v$ .*

1. *Every complex eigenvalue  $\mu$  of  $A$  satisfies  $|\mu| \leq \lambda$ .*
2. *Let  $N_k = \sum_{i,j} (A^k)_{ij}$  be the number of paths of length  $k$  in the graph of  $A$ . There are constants  $c, C > 0$  with  $c\lambda^k \leq N_k \leq C\lambda^k$  for all  $k$ . If  $A$  is primitive, then  $\lambda > 0$  and  $\frac{1}{k} \log N_k \rightarrow \log \lambda$ .*

*Proof.* (1) Let  $Aw = \mu w$  with  $w \neq 0$ , and choose  $i$  with  $|w_i|/v_i$  maximal; call this maximum  $t > 0$ . Then  $|w_j| \leq tv_j$  for all  $j$  and

$$|\mu| |w_i| = \left| \sum_j A_{ij} w_j \right| \leq \sum_j A_{ij} |w_j| \leq t \sum_j A_{ij} v_j = t \lambda v_i = \lambda |w_i|.$$

Since  $|w_i| > 0$ , this gives  $|\mu| \leq \lambda$ .

(2) From  $A^k v = \lambda^k v$  we get  $\sum_{i,j} (A^k)_{ij} v_j = \lambda^k S$  with  $S = \sum_i v_i$ . With  $a = \min v_j$  and  $b = \max v_j$  this gives  $a N_k \leq \lambda^k S \leq b N_k$ , so  $c = S/b$  and  $C = S/a$  work. Clearly  $\lambda \geq 0$ . If  $A^k$  is positive for some  $k \geq 1$ , then  $\lambda^k v_1 = \sum_j (A^k)_{1j} v_j > 0$ , so  $\lambda > 0$ . Taking logarithms in the bounds and dividing by  $k$  gives the limit.  $\square$

So a number realised in size  $n$  in our sense is the spectral radius of the matrix and the exponential of the entropy of  $X_A$ .

**Lemma 2.** *Let  $B$  be an  $n \times n$  matrix over  $\mathbb{Z}_{\geq 0}$  and let the real number  $\lambda$  be an eigenvalue of  $B$ . Then the minimal polynomial  $p$  of  $\lambda$  over  $\mathbb{Q}$  divides the characteristic polynomial  $\chi_B$ . In particular  $d(\lambda) \leq n$ . If  $d(\lambda) = n$ , then  $\chi_B = p$ , and the coefficient of  $X^{n-1}$  in  $p$  is  $-\text{tr } B \leq 0$ .*

*Proof.*  $\chi_B$  has rational coefficients and  $\chi_B(\lambda) = \det(\lambda I - B) = 0$ , so  $p$  divides  $\chi_B$ . Both are monic, and  $\chi_B$  has degree  $n$ . If  $p$  also has degree  $n$ , they are equal. The coefficient of  $X^{n-1}$  in  $\chi_B$  is  $-\text{tr } B$ , and the trace of a matrix with nonnegative entries is nonnegative.  $\square$

**Lemma 3.** *The polynomials  $p_1 = X^3 - X - 1$  and  $p_2 = X^3 + X^2 - 3X - 4$  are irreducible over  $\mathbb{Q}$ .*

*Proof.* A reducible cubic over a field has a root in that field. A rational root  $r = a/b$  in lowest terms of a monic cubic with integer coefficients has  $b \mid a^3$ , hence  $b = 1$ , and then  $r$  divides the constant term. For  $p_1$  the candidates are  $\pm 1$ , and  $p_1(1) = p_1(-1) = -1$ . For  $p_2$  the candidates are  $\pm 1, \pm 2, \pm 4$ , and

$$p_2(1) = -5, \quad p_2(-1) = -1, \quad p_2(2) = 2, \quad p_2(-2) = -2, \quad p_2(4) = 64, \quad p_2(-4) = -40.$$

$\square$

*Proof of Theorem 1. The roots.* We have  $p_1(5/4) = -19/64 < 0 < 5 = p_1(2)$  and  $p_2(9/5) = -41/125 < 0 < 2 = p_2(2)$ . By the intermediate value theorem there are real numbers  $x \in [5/4, 2]$  and  $y \in [9/5, 2]$  with  $p_1(x) = 0$  and  $p_2(y) = 0$ . By Lemma 3,  $p_1$  and  $p_2$  are their minimal polynomials, so  $d(x) = d(y) = 3$ , and both are algebraic integers.

*Perron numbers.* Dividing  $p_1$  by  $X - x$  shows that the other roots  $\mu$  of  $p_1$  satisfy  $\mu^2 + x\mu + x^2 - 1 = 0$ . The discriminant is  $4 - 3x^2 < 0$  because  $x \geq 5/4$ . So the two roots are complex conjugates with  $|\mu|^2 = x^2 - 1 < x^2$ . Likewise the other roots of  $p_2$  satisfy  $\mu^2 + (y + 1)\mu + y^2 + y - 3 = 0$ , with discriminant  $13 - 2y - 3y^2 < 0$  because  $y \geq 9/5$ , and  $|\mu|^2 = y^2 + y - 3 < y^2$  because  $y < 3$ . Hence  $x$  and  $y$  are Perron numbers.

*Realisations.* Let

$$A_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \quad A_4 = \begin{pmatrix} 0 & 0 & 1 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Both are primitive:

$$A_3^5 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 2 \end{pmatrix}, \quad A_4^5 = \begin{pmatrix} 6 & 1 & 12 & 8 \\ 12 & 6 & 9 & 2 \\ 1 & 4 & 6 & 8 \\ 4 & 4 & 1 & 2 \end{pmatrix}.$$

The vector  $u = (1, x, x^2)$  is positive and  $A_3 u = (x, x^2, 1 + x) = xu$ , because  $x^3 = x + 1$ . The vector  $w = (y, y^3 - 2y, y^2 - 2, 1)$  is positive, because  $y^2 \geq 81/25 > 2$ . We check  $A_4 w = yw$  row by row:

$$\begin{aligned} w_3 + 2w_4 &= y^2 = y w_1, & w_2 &= y(y^2 - 2) = y w_3, \\ w_1 + 2w_3 &= 2y^2 + y - 4 = y w_2, & w_1 &= y = y w_4. \end{aligned}$$

The identity  $w_1 + 2w_3 = y w_2$  uses  $y^3 = -y^2 + 3y + 4$ , which gives  $y^4 = 4y^2 + y - 4$  and so  $y w_2 = y^4 - 2y^2 = 2y^2 + y - 4$ . Hence  $x$  is realised in size 3 and  $y$  is realised in size 4.

*Lower bounds.* By Lemma 2, neither  $x$  nor  $y$  is an eigenvalue of a matrix of size 1 or 2 over  $\mathbb{Z}_{\geq 0}$ . So  $m(x) = 3$ . If  $y$  were an eigenvalue of a  $3 \times 3$  matrix  $B$  over  $\mathbb{Z}_{\geq 0}$ , Lemma 2 would give  $\chi_B = p_2$  and  $\text{tr } B = -1$ , which is impossible. So  $m(y) = 4$ .  $\square$

*Remark.* (i) The characteristic polynomial of  $A_4$  is  $(X - 1)(X^3 + X^2 - 3X - 4)$ ; the extra eigenvalue 1 makes the trace zero.

(ii) The lower bounds do not use primitivity, and  $A_3, A_4$  are primitive. So (C1) fails in the same way if “realising” allows irreducible or arbitrary nonnegative integer matrices. If only 0–1 matrices are allowed,  $A_3$  is still admissible,  $y$  still needs size at least 4, and  $y$  is realised by the 0–1 edge matrix of the graph of  $A_4$ ; so  $m(x) \neq m(y)$  under that reading too. The statement about 0–1 matrices is not formalised.

(iii) The converse implication in (C2) also fails. The polynomial  $X^3 - X^2 + X - 3$  is irreducible, its real root  $z \approx 1.575$  is a Perron number, and its coefficients alternate in sign. A  $3 \times 3$  matrix  $B$  over  $\mathbb{Z}_{\geq 0}$  with this characteristic polynomial would have trace 1, so exactly one diagonal entry equal to 1 and the others 0; then the sum of its principal  $2 \times 2$  minors is  $-\sum_{i < j} B_{ij} B_{ji} \leq 0$ , but this sum is the coefficient  $+1$  of  $X$ . By Lind’s theorem  $z$  is realised in some size, so  $m(z) > 3 = d(z)$ . This remark is not formalised and is not needed for the disproof.

## Formal verification and scope

The Lean file works with Mathlib’s `Matrix`, `minpoly`, `Matrix.charpoly` and `Polynomial`. For a matrix  $A$  with natural number entries indexed by `Fin n`:

- `IsPrimitive A`: some power  $A^k$  with  $k \geq 1$  has all entries positive.
- `IsPerronEigenvalue A lam`: there is a real vector with positive entries and  $Av = \lambda v$ .

- **Realizable lam n**:  $n \geq 1$  and some primitive  $n \times n$  matrix has Perron eigenvalue  $\lambda$ .
- **IsMinSize lam m**:  $\lambda$  is realisable in size  $m$  and in no smaller size.
- **algDegree lam**: the degree of Mathlib’s minimal polynomial `minpoly` of  $\lambda$  over  $\mathbb{Q}$ .
- **IsPerronNumber lam**:  $\lambda$  is integral over  $\mathbb{Z}$ ,  $\lambda \geq 1$ , and every complex root  $\mu \neq \lambda$  of the minimal polynomial has  $|\mu| < \lambda$ .

Lemma 1 is proved as

`norm_le_of_isPerronEigenvalue, pathCount_bounds, hasEntropy_log.`

The last one states that  $\frac{1}{k} \log N_k$  converges to  $\log \lambda$  for a primitive matrix with Perron eigenvalue  $\lambda$ , where  $N_k$  is `pathCount A k`. Lemma 2 is `minpoly_dvd_charpoly, algDegree_le` and `not_isEigenvalue_three`. Lemma 3 is `p1_irreducible` and `p2_irreducible`. The roots are obtained from the intermediate value theorem. Theorem 1 is

`two_cubic_perron_numbers`

which states the existence of real  $x, y$  that are Perron numbers of algebraic degree 3 with `IsMinSize x 3` and `IsMinSize y 4`, and such that the minimal polynomial of  $x$  is not weakly sign-alternating.

The refuted statements are the following propositions.

- **ClaimedDegreeLaw**: there is  $f : \mathbb{N} \rightarrow \mathbb{N}$  such that every Perron number  $\lambda$  has minimal size  $f(d(\lambda))$ . Its negation is `conjecture_false`.
- **ClaimedDegreeLawRealized**: the same for every  $\lambda$  that is realised in some size. Its negation is `conjecture_false_realized`. This version does not depend on the definition of Perron numbers.
- **ClaimedSignCriterion**: for every Perron number,  $m(\lambda) = d(\lambda)$  if and only if the minimal polynomial is weakly sign-alternating, meaning that no two adjacent coefficients have the same strict sign. Its negation is `sign_criterion_false`. Every polynomial with alternating coefficient signs, with or without zero coefficients allowed, is weakly sign-alternating, so the refuted implication fails for those notions too.

No `sorry`, `native_decide` or custom axiom occurs.

Not formalised: Lind’s theorem (it is not used; it only explains why the two quantifications describe the same set of numbers); the Perron–Frobenius theorem that every primitive matrix has a positive eigenvector (it is not used: realisability is defined through the eigenvector, and the lower bounds are proved for every real eigenvalue of every nonnegative integer matrix); the equivalence of primitivity and mixing and the identification of  $\lim \frac{1}{k} \log N_k$  with topological entropy, which are standard facts used here as definitions; and the three items of the Remark. The source does not define “sign-alternating”; under our reading of (C2) one direction is refuted in Lean and the other on paper.

## Source and reproduction

The exact bilingual source is included byte-for-byte as `SOURCE.md`. Its repository path is

`conjectures/00000009611.md`.

The project pins Lean 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

From `lean/`, run `lake build`, then

```
lake env lean -DwarningAsError=true Main.lean.
```

The included public-Git manifest locks all dependency revisions. The `verification/` directory records the fresh-build logs, theorem-axiom reports, artifact hashes, review and final PDF inspection. The matrix  $A_4$  was found by a computer search; its properties are proved above and in Lean, and no numerical computation is part of the proof.